

# FDTT 704: Introduction to R

Dr. Mutua Kilai, PhD

September-December 2026

Department of Pure and Applied Sciences

# Introduction

- This lecture will heavily borrow from the book Hands-on Programming with R by Garrett Grolmund
- The book was wrote for non-programmers to provide a friendly introduction to the R language.
- You'll learn how to load data, assemble and disassemble data objects, navigate R's environment system, write your own functions, and use all of R's programming tools.
- Throughout the book, you'll use your new found skills to solve practical data science problems.

# Obtaining R and RStudio

- R can be downloaded from the website: <http://cran.r-project.org/>.
- See additional online notes which give a very detailed description on downloading and installing R (and Bioconductor).
- R is available for all platforms: Unix/Linux, Windows and Mac.
- In this course we will call R using RStudio which provides a richer interface and nicer editor of R code. RStudio can be obtained from [www.rstudio.org](http://www.rstudio.org)).
- RStudio is available as a desktop program for all platforms Unix/Linux, Windows and Mac.

# The Defaults R interface

- When you click the R icon, you will open the basic R interface
- On the menu bar there are the menus:
  - i. File- load script, load/save session
  - ii. Edit - Cut/Paste. GUI interfaces
  - iii. View
  - iv. Misc - stop computations, list/remove objects in the session
  - v. Packages - allows one to install new, update packages

# Default R Console

- The main window that you see in the R Console. This is where you type R commands.
- Within the console you can:
  - i. Type at the prompt symbol
  - ii. If the prompt instead show the continuation prompt is '+', it means the command you typed is incomplete and R is waiting for it to complete
  - iii. Use up and down arrow keys to scroll through previous commands. This is useful if you would like to repeat a previous command

# Default R Editor

- Within R, one can open an editor using the menu command *File* – *> Newscript*
- Can type commands in the editor
- Highlight the commands and type ctrl + R to submit the commands for R evaluation



# Cont'd

- Evaluation of the commands can be stopped by pressing the Esc key or the Stop button
- Saving (ctrl + s) and opening saved commands (ctrl + o)
- Can open a new editor by typing (ctrl + N)

# R shortcuts

- Ctrl + A: gets you to the beginning of the line
- Ctrl + E: gets you to the end
- Ctrl + K: wipes everything to the right of your cursor
- Esc: Kills the current moments
- Up Arrow: brings back last command you typed
- Down Arrow: brings back the next command



# RStudio Interface

- RStudio is a free and open source integrated development environment for R.
- On startup R Studio brings up a window with 3 or 4 panels.
- If you only see 3 panels, click on *File* – *>* *New* – *>* *NewRScript*.
- The first thing to notice, is that the bottom left panel “console” is the exact same as the standard R console.
- RStudio just loads your local version of R.

# Console

- RStudio has nice console features.
- Has auto suggestion feature
- Press F1, it will bring up a help document about the function in the help panel (right bottom)
- Press F2, it will show the source code for the function

# Editor

- The top left panel is an editor which can be used to edit R scripts (.R), plain text (.txt), html web files or Sweave (rnw) or markdown (md), the latter two of these can be converted to pdf files.

# Workspace history

- Workspace list the objects in the current R session. You can load, save or "Clear All" object for a workspace
- Note that under the workspace panel there is the option to Import Dataset
- The history panel lists all of the command that have been typed or input in the console.
- There are options to load, save, search or delete history
- One can easily repeat a command by highlighting one or more line(s) and sending these To Console

# Files, and Plots

- **Files** is a file browser, which allows you to create a new folder, rename a folder or delete a folder. Click on the triple dot icon (...) on the far right on the menu to browse folders. Under the More menu you can set your current working directory (more about that below). If you double click on a text, .R, Sweave or html file it will automatically open in the Editor
- The **Plots** windowd is plays plots generated in R. Simply type the following command into the Console window

# Packages and Help

- **Packages** list all of the packages installed in your computer. The packages with tick marked are those loaded in your current R session. Click on a package name to view help on that package. Note you can install packages or check for updates. You can also search for a package or search package descriptions using the search window.
- The **Help** menu provides an extensive R help. The arrows button go forward or back through recent help pages you have viewed. You can go home (house icon), print or open help in new window. You can search help, use the search window. Help can also be browsed through main menu bar at the open of the page



# Projects in RStudio

- RStudio provides an easy approach to managing projects. In the main menu there is a **Projects** menu which will create folder for your project and retain all data files and command history for your project. It is also possible to set up a backup subversion management control system for your code as RStudio will directly communicate with your SVN or github account.

 Important

*We will create project for our class this semester where we will store all our files, datasets and any other important information*

# R as a Calculator

- We can use R as a calculator as follows:

```
1 # addition
2
3 2+ 2
4
5 # subtraction
6
7 10-2
8
9 # multiplication
10
11 6*7
12
13 # Division
14
15 12/3
```



# Important Points

- Elementary commands: *expressions* are evaluated, printed and value lost; assignments evaluate expressions, passes value to a variable but not automatically printed
- Assignment operators are  $< -, =, \leftarrow >$
- R is case sensitive  $x$  and  $X$  are two different names

# Cont'd

- Missing values are represented by NA
- Comments can be put anywhere. To comment text, insert a hashmark #. Everything following it to end of the line is commented out (ignored, not evaluated).

# Basic Operators

# Comparison Operators

- Equal =
- not equal: !=
- greater/less than  $>$ ,  $<$
- greater/less than or equal  $>=$ ,  $<=$

# Logical Operator

- AND & Returns TRUE if both comparisons return TRUE
- OR: Returns TRUE where at least one comparison returns TRUE
- NOT ! The sign ! returns the negation (opposite) of a logical vector

# Data sets in R

- Both the R core installation and contributed R package contain data sets, which are useful example data when learning R. To list all available data sets:

```
1 data()
```



# R Packages

- By default, R is packaged with a small number of essential packages, however as we saw there are many contributed R packages.
- Some packages are loaded by default with every R session. The libraries included in the Table are loaded on the R startup.

# Cont'd

| package   | description                                |
|-----------|--------------------------------------------|
| base      | Base R Functions                           |
| datasets  | Base R datasets                            |
| grDevices | Graphics device for base and grid graphics |
| graphics  | R functions for base graphs                |
| methods   | Formally defined methods and classes       |
| stats     | R statistical functions                    |
| utils     | R utility functions                        |



# Cont'd

- To see which packages are currently loaded, use

```
1 search()  
2 sessionInfo()
```

- To see which packages are installed on your computer, issue command

```
1 library()
```

# Installing New Libraries

- There are several thousand R packages and >500 Bioconductor packages (also called libraries) available. These are not installed by default, so we have to select and install additional packages that will be of use to us.

```
1 # installing and updating libraries
2
3 install.packages("lme4")
4 update.packages("lme4")
```

# Objects in R

- Everything in R is an object. Every object has a class
- R creates and manipulates objects: variables, matrices, strings, functions, etc. objects are stored by name during an R session.
- During a R session, you may create many objects, if you wish to list the objects you have created in the current session use the command

```
1 objects()  
2 ls()
```

# Types of R Objects

- Objects can be thought of as a container which holds data or a function. The most basic form of data is a single element, such as a single numeric or a character string.

# Vectors

- A **vector**: An ordered collection of numerical character, complex or logical objects.
- Vectors are collection of *atomic* components of modes

```
1 # numeric
2 vec1 <- 1:10
3
4 # character
5 vec2 <- LETTERS[1:10]
6
7 # logical
8 vec3 <- vec2=="D"
```

# Matrix

- A matrix is a multidimensional collection of data entries of the same type. Matrices have two dimensions. It has row names and col names.

```
1 mat1 <-matrix(vec1, ncol =2, nrow=5)
2 # dimension
3 dim(mat1)
4
5 colnames(mat1) <-c("A", "B")
6 rownames(mat1) <-paste("N",1:5, sep="")
```



# List

- A list is an ordered collection of objects that can be of different modes e.g numeric vector, array etc

```
1 a <-20
2
3 newList1 <-list(a,vec1, mat1)
4
5 newList1 <-list(a=a, myVec=vec1,mat=mat1)
```

# Factor

- A factor is a vector of categorical variables, it can be ordered or unordered.

```
1 charVec <- rep(LETTERS[1:3], 10)
2 table(charVec) # Tabulate charVec
3 fac1 <- factor(charVec)
4 # attributes
5
6 attributes(fac1)
7
8 # levels
9
10 levels(fac1)
```



# Array

- An array in R can have one, two or more dimensions.

```
1 array(1:24, dim = c(2, 4, 3))
```

# Reading and Writing Data to and From R

- We shall see how to import and export data to and from R
- The most commonly used function for reading data is `read.table()`. It will read the data into R as a `data.frame`.

```
1 # importing delimited data
2 data <- read.table("Women.txt", sep = "\t", header = TRUE)
```

# using read.csv()

- is a derivative of read.table() which calls read.table() function with the following options so it reads a comma separated file:

```
1 read.csv(file,header=TRUE,sep="," ,quote="\\"",dec=".", fill=TRUE,comment.char="",...)  
2  
3 # Comma Delimited  
4 data2 <-read.csv("data.csv")
```

# Reading direct from website

```
1 myURL <"http://bcb.dfci.harvard.edu/~aedin/courses/Bioconductor/Women.txt"  
2 read.table(myURL, header = TRUE)[1:2, ]
```

# Reading data from Excel to R

```
1 library(xlsx)
2 ww <- read.xlsx(file = "Wdata.xlsx", sheetIndex = 1)
```

# Important References Materials

1. <https://rstudio-education.github.io/hopr/basics.html>
2. <https://ds.dfc.harvard.edu/~aedin/courses/R/CDC/Intro2R.pdf>
3. <https://r4ds.had.co.nz/>

## ⚠ Important

This was just an introductory session to enable us set a basis for using R for the classes.  
For more info then you read on your own the materials shared here.